



Activity budgets shape parent–offspring conflict expression: Evidence from mantled howler monkeys

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Abstract Parent–offspring conflict (POC) is a cornerstone of evolutionary theory, yet empirical tests have concentrated on cercopithecine primates, limiting our understanding of how ecological constraints shape conflict expression. We characterized POC behavioral dynamics in mantled howler monkeys (*Alouatta palliata*), testing whether predictions derived from energy-maximizing species generalize to time-minimizing taxa. We observed 16 mother-infant dyads across four groups in Los Tuxtlas, Mexico, recording maternal rejection, offspring solicitation, and spatial relationships across lactation stages and maternal activities. Maternal rejection increased significantly between early and middle lactation during foraging and inactivity, but was highest during inactivity in late lactation, contrasting with predictions that foraging should be the primary conflict context. Offspring solicitation behaviors (pursuing and following) increased between early and middle lactation but plateaued rather than escalating further. Responsibility for proximity maintenance rested primarily with infants throughout development, reflecting the permissive maternal style characteristic of this species. Time in contact decreased while distance increased across lactation stages regardless of maternal activity. These findings extend POC research to an understudied platyrrhine species and demonstrate that conflict concentrates during whichever activity carries highest opportunity costs for mothers—inactivity rather than foraging in this time-minimizing species—refining the timing hypothesis for application across diverse foraging strategies. Our results underscore the necessity of incorporating species-specific ecology into behavioral predictions derived from evolutionary theory.

Significance statement Evolutionary theory predicts that parents and offspring will disagree over how much care parents should provide, a conflict rooted in differences in genetic interests. However, most tests of this theory come from a narrow range of species with similar lifestyles, leaving unclear whether predictions apply broadly. By studying howler monkeys, primates whose time-minimizing foraging strategy results in activity budgets dominated by rest rather than active foraging, we show that mother-offspring conflict peaks during rest periods, not during feeding as observed in more active, energy-maximizing species. This finding suggests that conflict concentrates whenever maternal activities are most critical to the mother's own well-being, whether that involves acquiring food or processing it. Our results demonstrate that applying evolutionary theory across ecologically diverse species uncovers both universal principles and the need to account for each species' unique way of life.

Keywords Parent–offspring conflict · Maternal investment · Primate development · Maternal rejection · *Alouatta palliata* · Maternal activities

Introduction

Parent–offspring conflict (POC) is a cornerstone of evolutionary theory that explains how genetic relatedness shapes family interactions across taxa (Trivers 1974). The conflict arises because parents and offspring have different optimal levels of parental investment: parents, equally related to all current and future offspring ($r=0.5$), are selected to distribute investment across their reproductive lifespan, whereas each offspring, more related to itself ($r=1.0$) than to siblings

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($r=0.5$), is selected to secure more than its equitable share (Trivers 1974; Mock and Parker 1997). This asymmetry creates a zone of conflict when the cost-to-benefit ratio of continued investment falls between 1 and 0.5; the range where parental and offspring fitness optima diverge (Trivers 1974). Understanding how this fundamental conflict manifests behaviorally remains central to evolutionary biology, with implications for life history evolution, signaling theory, and social behavior.

Parent–offspring communication can remain evolutionarily stable despite conflicting interests (Godfray 1991, 1995). Under the honest signaling framework, offspring solicitation is costly but conveys reliable information about need, enabling adaptive parental responses (Godfray 1991; Bateson 1994). This contrasts with scramble competition models predicting escalating offspring demands regardless of condition (Parker and Macnair 1979). Distinguishing between these frameworks requires examining whether offspring behavior tracks developmental need or simply increases until independence, a question best addressed through detailed longitudinal observation of parent–offspring interactions.

Behaviorally, POC manifests through offspring solicitation—such as physical approaches, contact-seeking, and vocalizations—and parental rejection ranging from nipple denial to physical repulsion (Kilner and Johnstone 1997; Maestripieri 2002; Pavé et al. 2010). Importantly, conflict expression is not uniform but varies with ecological context. Altmann's (1980) timing hypothesis predicts that conflict should peak during activities like foraging in early infancy, when parental care is most costly, thereby conditioning offspring about appropriate times to solicit investment. This prediction has found support in species with high foraging demands, such as gelada baboons (*Theropithecus gelada*; Barrett et al. 1995), but remains untested in species with fundamentally different activity budgets and foraging strategies.

A key distinction in foraging ecology is between energy-maximizing and time-minimizing strategies (Hixon 1982; Knott and DiGiorgio 2024). Energy maximizers increase fitness by maximizing the rate of energy intake and spend a large proportion of their time actively foraging, whereas time minimizers meet a set energy requirement and then allocate remaining time to non-foraging activities critical to fitness (Hixon 1982). This distinction has important implications for POC theory: if conflict peaks during whichever activity imposes the greatest opportunity costs on mothers, then the primary arena for conflict should differ between energy maximizers—where foraging time is at a premium—and time minimizers—where non-foraging activities such as rest and digestion may be more critical.

Life history theory predicts that POC dynamics should vary systematically across species with different reproductive strategies (Charnov and Berrigan 1993; Stearns 1998). Species with slow life histories and extended parental care periods provide prolonged opportunities for conflict to unfold, potentially revealing dynamics obscured in species with rapid development (Lee and Kappeler 2003; Raguet-Schofield and Pavé 2015). Yet empirical tests of POC theory have concentrated heavily on cercopithecine primates characterized by active lifestyles and substantial foraging time (reviewed in Maestripieri 2002). Although work on platyrrhine taxa has begun expanding the taxonomic scope of POC research (Pavé et al. 2010, 2015; Arbaiza-Bayona et al. 2022), our ability to evaluate how foraging strategies shape conflict expression remains limited.

Howler monkeys (*Alouatta* spp.) offer a valuable opportunity to test POC theory under distinct ecological conditions. Their folivorous-frugivorous diet necessitates extended gut passage time, resulting in a time-minimizing foraging strategy in which activity budgets are dominated by inactivity (up to 80% of daily time; Milton 1980; Di Fiore and Campbell 2007). They allocate less than 5% of the daily time budget to social interactions, considerably less than cercopithecines and atelines (Di Fiore and Campbell 2007). In mantled howler monkeys (*A. palliata*), infants are nursed exclusively until approximately 5 months, begin consuming solid food thereafter, and are weaned between 12 and 14 months of age (Balcells and Veà 2009; Rangel-Negrín et al. 2025). Females reach first reproduction at approximately 43 months with interbirth intervals averaging 22 months (Glander 1980; Dias et al. 2023). This ecological and life history profile creates predictions that diverge from those for more active, energy-maximizing species: if the timing hypothesis reflects general principles about cost-sensitive conflict expression, then inactivity—instead of foraging—may constitute the primary arena for mother–offspring negotiation in time-minimizing species.

Research on black and gold howler monkeys (*A. caraya*) provides the most detailed precedent for POC in the genus. In a preliminary study, Pavé et al. (2010) documented maternal rejection of suckling and transport in five immatures, finding that rejection increased with offspring age and that all suckling attempts were rejected following the birth of new siblings; notably, they suggested that mothers may interpret infant distress calls as honest signals of need, providing an early application of honest signaling theory to *Alouatta*. Pavé et al. (2015) subsequently examined 37 mother–offspring dyads across two populations differing markedly in population density, home range overlap, and intergroup encounter rates, finding that rejection rates were highest between months 3 and 7, coinciding with the transition from exclusive nursing to mixed feeding. Importantly,

the patterns of mother–offspring conflict were remarkably consistent across the two demographically distinct sites, suggesting that proximate mechanisms of conflict expression may be robust to variation in social and demographic context (Pavé et al. 2015). Life history analyses further indicate that *Alouatta* females may have greater flexibility in maternal investment than other atelines, given their higher potential lifetime reproductive output and shorter interbirth intervals (Raguette-Schofield and Pavé 2015), which may shape how mothers negotiate the trade-off between current and future reproduction. However, these studies did not examine whether conflict expression varies across maternal activities, leaving untested whether the activity-specific predictions of the timing hypothesis apply to howler monkeys.

Here, we characterize POC behavioral dynamics in mantled howler monkeys, providing the first comprehensive test of how conflict varies across maternal activities and developmental stages in this species. We leverage recent advances in understanding mantled howler maternal behavior, including documentation of flexible maternal care styles (Czerwinski et al. 2025a) and patterns of infant handling (Czerwinski et al. 2025b), to interpret conflict dynamics within their social context. We test four predictions: (P1) following POC theory (Trivers 1974), maternal rejection increases through lactation as the cost-to-benefit ratio of continued investment shifts; (P2) following the timing hypothesis (Altmann 1980), rejection is higher during foraging than during other activities; (P3) following honest signaling models (Godfray 1991, 1995), offspring solicitation behaviors increase between early and middle lactation as locomotor independence develops and nutritional need is greatest, then decrease in late lactation as infants approach nutritional independence; and (P4) following predictions from maternal investment theory (Trivers 1974; Maestripieri 2002), responsibility for proximity maintenance shifts from mothers to offspring across development. By testing these predictions in a time-minimizing species with an activity budget fundamentally different from those

previously studied, we evaluate whether POC theory's predictions are robust across diverse foraging strategies or require modification to accommodate species-specific ecological constraints.

Methods

Study sites and subjects

We conducted this study at Los Tuxtlas in southeast Veracruz, Mexico. This region is composed of several types of vegetation, including high evergreen rainforest and mid-sub-evergreen rainforest, as well as areas of pastureland. There is a rainy season from June to December and a dry season between January and May, with annual rainfall varying between 1500 and 4500 mm. Ambient temperatures range between 22 and 26 °C (Vázquez-Torres et al. 2010).

The study groups lived in three different forest fragments (Table 1), and we followed 16 mother–infant dyads from four groups. We identified study subjects via natural markings, such as patches of yellow fur in their tails and feet, scars, and facial features.

Behavioral sampling

We conducted fieldwork from September 2021 to September 2023, collecting 850 h of data. We followed study groups for 8 h per day, during which a single observer used focal animal sampling with continuous recording to study mothers with infants. We performed field observations weekly, four days a week, usually from 6:00–7:00 to 16:00. Due to the challenges associated with finding and following the focal individuals, each day we tried to sample mothers for as long as possible, resulting in variation in focal sample duration (mean $\pm$ SD = 3.8 ± 1.5 h, range = 1.1–7.0 h). We sampled each mother–offspring dyad on a mean of 4.1 ± 1.3 days per lactation stage, with a minimum of 5 h of focal observations

Table 1 Study locations and sampling effort for 16 mantled howler monkey mother–infant dyads studied at Los Tuxtlas

Attribute	Group			
	Group 1	Group 2	Cerro del Borrego	Balzapote
Location (UTM)	2,040,592.64 N 283,418.13 E	2,040,000.09 N 283,265.46 E	2,062,417.38 W 279,494.73 E	2,058,650.73 W 282,352.91 E
Habitat size (ha)	100	100	78	8
Group size	10	15–19	23	8–12
Number of sampled mother–infant dyads	1	6	2	7
Dyads sampled in early lactation ^a	1 (5.4)	6 (113.5)	1 (13.8)	6 (133.0)
Dyads sampled in middle lactation	1 (5.8)	5 (148.7)	1 (12.6)	5 (129.1)
Dyads sampled in late lactation	0	4 (103.2)	0	3 (78.1)

^aNumbers in parenthesis are the number of sampling hours

per dyad per stage. It was not possible to record data blind because our study involved focal animals in the field.

We focused on mothers with associated infants younger than 15 months of age, i.e., non-weaned (Balcells and Veà 2009). We classified infants into three developmental stages corresponding to lactation periods: early lactation (infant 1: 0–3 months), middle lactation (infant 2: 4–8 months), and late lactation (infant 3: 9–14 months) (Balcells and Veà 2009; Czerwinski et al. 2025a). We determined offspring age through long-term demographic records ($n=3$), observation of births ($n=10$), or estimation based on physical characteristics such as coat color and body size ($n=3$; Balcells and Veà 2009). Not all dyads were observed across all lactation stages; the dataset comprises 14 dyads in early lactation, 11 in middle lactation, and 7 in late lactation.

We recorded behaviors of mothers and offspring using focal-animal sampling with continuous recording for rejection (P1 and P2), pursuing (P3), following (P3), approaching (P4), leaving (P4), making contact (P4), and breaking contact (P4; Table S1). We also recorded the duration of maternal activities as foraging, inactive, and moving (Table S1). During focal samples we used instantaneous recordings at 5 min intervals to study the spatial relationships between mothers and offspring as contact (dorsal, ventral, or other body area), proximity (1 to 5 m), and away (>5 m).

Data analysis

We fitted generalized linear mixed models (GLMMs) using the `glmmTMB` package (Brooks et al. 2017) and Dirichlet regression models using the `DirichletReg` package (Maier 2014) in R version 4.4.0 (R Core Team 2025). For GLMMs, we assessed model fit using residual diagnostics from the DHARMA package (Hartig 2022), including tests for overdispersion and zero-inflation. When these conditions were detected, we applied appropriate distributional families and zero-inflation parameters. For all statistical models, we assessed the significance of predictor variables using Type II or Type III Analysis of Deviance (Anova function from the `car` package; Fox and Weisberg 2018), depending on the presence of interaction terms. We used the `emmeans` package (Lenth and Piaskowski 2025) for post-hoc pairwise comparisons with Tukey adjustment for multiple comparisons.

Spatial relationships

To characterize developmental changes in spatial relationships between mothers and offspring, we modeled the proportion of time spent at different distances per focal sample using Dirichlet regression (`DirichletReg` package; Maier 2014), which is appropriate for compositional data where proportions must sum to one. We aggregated proximity

data by dyad, observation day, lactation stage, and maternal activity, calculating the proportion of time spent in each distance category (contact, proximity, and away). We fitted a Dirichlet regression model with maternal activity, lactation stage, and their interaction as predictors. We used parametric bootstrapping (2,000 iterations) to obtain 95% confidence intervals for predicted proportions and to conduct pairwise comparisons between factor levels, computing bootstrap-based p-values as the proportion of bootstrap samples where the absolute difference exceeded the observed difference.

Maternal rejection (P1, P2)

To test whether maternal rejection increased through lactation (P1) and was higher during foraging than during other activities (P2), we fitted a GLMM with a negative binomial distribution (`nbinom2` family) to account for overdispersion. The response variable was the frequency of rejection per maternal activity per focal sample (Table S2). The model included lactation stage (early, middle, late), maternal activity (foraging, moving, inactive), and their interaction as fixed effects. We included a nested random effect structure with observation day nested within mother-infant dyad identity to account for repeated measures and between-dyad variation. We included an offset term for the log-transformed duration of each activity to model rejection as a rate. Based on model diagnostics, we included a zero-inflation parameter to account for excess zeros in the rejection counts.

Offspring pursuing and following behaviors (P3)

To test whether offspring increased solicitation behaviors between early and middle lactation and decreased these behaviors between middle and late lactation (P3), we fitted separate GLMMs for the frequency of pursuing and following per focal sample using a negative binomial distribution. Both models included lactation stage (early, middle, late) as a fixed effect and mother-infant dyad as a random effect. We included offset terms for the log-transformed duration of relevant maternal activities to model behaviors as rates: total moving time for pursuing (as pursuing occurs when offspring attempt to catch up with moving mothers) and total foraging time for following (as following occurs when offspring trail foraging mothers).

Responsibility for proximity maintenance (P4)

To test whether responsibility for maintaining and breaking contact shifted between mother and offspring across lactation stages, and whether these patterns varied by maternal activity context, we analyzed the frequency of approaching,

leaving, making contact, and breaking contact per focal sample. For each behavior, we fitted separate generalized linear mixed models (GLMMs) within each activity context (foraging, inactivity, moving). We fitted each model with a negative binomial distribution with actor (mother/infant), lactation stage (early, middle, late), and their interaction as fixed effects. We included a nested random effect structure to account for repeated observations within days and dyads, and an offset term for the log-transformed activity-specific observation time to model behaviors as rates. One model (leaving during moving) failed to converge and was excluded from further analysis.

Results

Developmental changes in spatial relationships

Spatial relationships between mothers and infants changed markedly across lactation stages (Fig. 1). During early lactation, mothers and offspring spent the majority of their time in contact regardless of maternal activity: 68% during foraging, 83% during inactivity, and 91% during moving. Time spent away was correspondingly low (7–14% across activities). By middle lactation, time in contact had decreased substantially across all activities. During foraging, contact time dropped to 20% while time away increased to 49%. During inactivity, contact time decreased to 51% and time away increased to 27%. During moving, contact time fell to

39% and time away rose to 44%. These changes continued into late lactation. During foraging, dyads spent only 11% of time in contact and 67% away. During inactivity, contact time decreased further to 35% while time away reached 42%. During moving, contact time was lowest at 13%, with 70% of time spent away. Bootstrap contrasts confirmed that changes across lactation stages were significant for all activity contexts. Time in contact decreased significantly between early and middle lactation (all bootstrap $p < 0.52$, 95% CIs excluding zero; Table S3) and between early and late lactation (all $p < 0.54$). Time spent away showed the inverse pattern, increasing significantly across lactation stages within all activities.

Maternal rejection across lactation stages and activities (P1, P2)

We found significant effects of lactation stage ($\chi^2 = 59.3$, $df = 2$, $p < 0.001$), maternal activity ($\chi^2 = 6.9$, $df = 2$, $p = 0.031$), and their interaction ($\chi^2 = 26.7$, $df = 4$, $p < 0.001$) on the frequency of maternal rejection (Fig. 2, Table S4).

The pattern of rejection across lactation stages differed by maternal activity (Tables S5, S6). During foraging, rejection significantly increased between early and middle lactation ($z = -4.4$, $p < 0.001$) and then significantly decreased between middle and late lactation ($z = -3.1$, $p = 0.006$), with no significant difference between early and late lactation ($z = -1.3$, $p = 0.388$). During inactivity, rejection increased continuously throughout lactation, with significant increases

Fig. 1 Developmental changes in spatial relationships between mothers and infants across lactation stages during different maternal activities in 16 mantled howler monkey mother-infant dyads (14 early, 11 middle, 7 late lactation) studied at Los Tuxtlas. Bars show predicted proportions of time spent in contact (orange), proximity (0–5 m; blue), and away (> 5 m; green) from Dirichlet regression (N=411 observations)

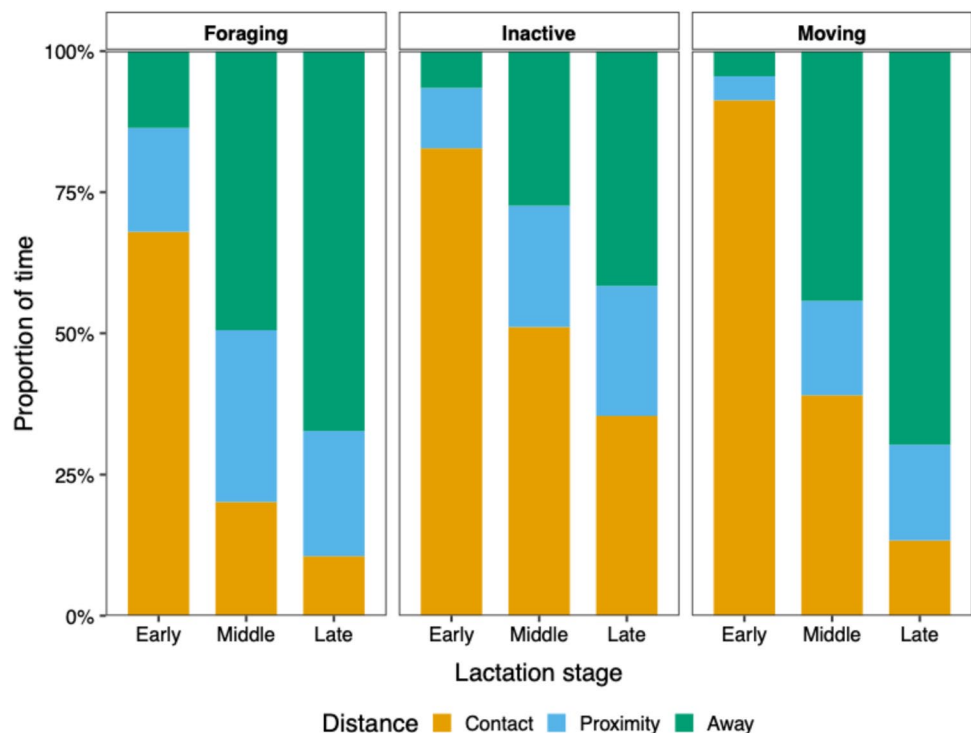
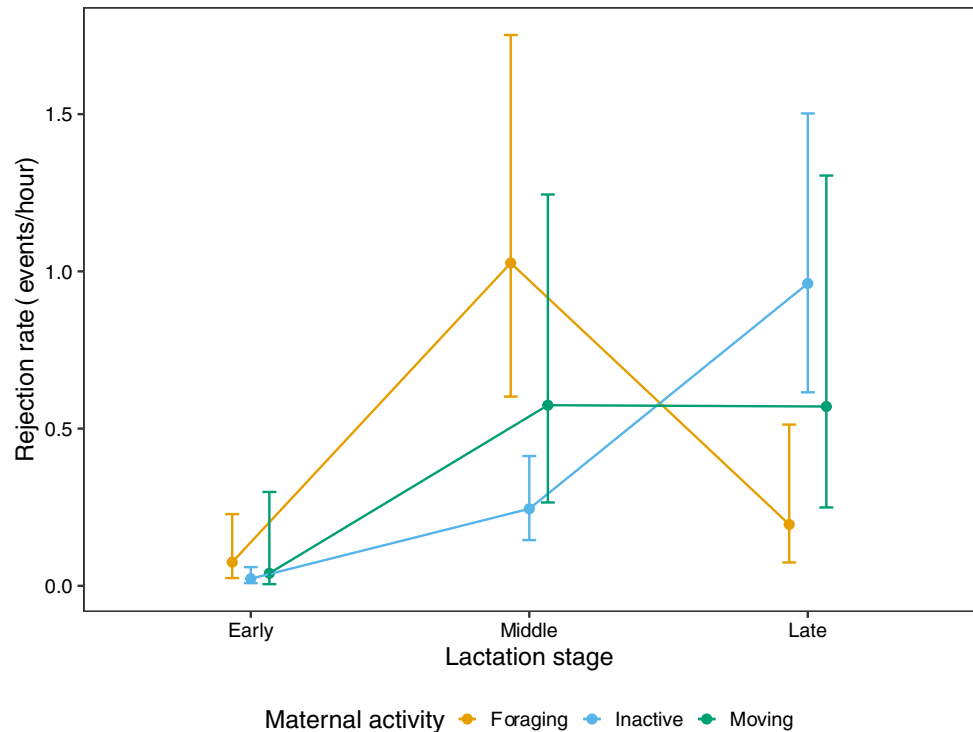


Fig. 2 Estimated marginal mean rates of maternal rejection (events/hour) across lactation stages for each maternal activity in 16 mantled howler monkey mother-infant dyads (14 early, 11 middle, 7 late lactation) studied at Los Tuxtlas. Points represent model predictions from a generalized linear mixed model with negative binomial distribution, back-transformed to the response scale; error bars indicate 95% confidence intervals (N=411 focal samples)



between early and middle lactation ($z = -4.5$, $p < 0.001$) and between middle and late lactation ($z = 4.1$, $p < 0.001$). During moving, rejection significantly increased between early and middle lactation ($z = -2.5$, $p = 0.037$) and remained stable between middle and late lactation ($z = -0.01$, $p = 0.999$).

Rejection was not consistently higher during foraging across all stages. During early lactation, there were no significant differences in rejection rates among activities (all $p > 0.21$). During middle lactation, rejection was significantly higher during foraging than during inactivity ($z = 4.6$, $p < 0.001$), with no significant differences between foraging and moving ($z = 1.4$, $p = 0.356$) or between inactivity and moving ($z = -2.0$, $p = 0.109$). During late lactation, rejection was significantly higher during inactivity than during foraging ($z = -3.1$, $p = 0.006$), with no significant differences between foraging and moving ($z = -1.7$, $p = 0.190$) or between inactivity and moving ($z = 1.2$, $p = 0.481$).

Infant pursuing and following behaviors (P3)

Lactation stage significantly affected pursuing behavior ($\chi^2 = 44.6$, $df = 2$, $p < 0.001$; Table S7). Pursuing significantly increased between early and middle lactation ($z = -6.1$, $p < 0.001$; Table S8). However, there was no significant decrease between middle and late lactation ($z = 0.01$, $p = 1.000$); pursuing rates remained elevated and nearly identical between these stages (Fig. 3A). Similarly, lactation stage significantly affected following behavior ($\chi^2 = 20.3$, $df = 2$, $p < 0.001$). Following significantly increased between

early and middle lactation ($z = -4.5$, $p < 0.001$; Table S8), but there was no significant decrease between middle and late lactation ($z = -0.5$, $p = 0.866$; Fig. 3B).

Responsibility for proximity maintenance (P4)

Approaching

Actor and lactation effects on approaching varied by activity context (Table 2, S9, Fig. 4A). During foraging and inactivity, offspring displayed significantly higher approaching rates than mothers across all lactation stages (Table S13). During inactivity, offspring consistently approached more than mothers with no significant change across stages. During moving, approaching rates did not differ significantly between actors, although lactation had a strong effect on overall rates. No pairwise contrasts between mother and offspring were significant at any lactation stage during moving (all $p > 0.05$).

Leaving

Leaving behavior showed distinct patterns across activity contexts (Table 2, S10, Fig. 4B). During foraging, mothers and offspring left at similar rates regardless of lactation stage (Table S13). During inactivity, offspring left at significantly higher rates than mothers across all lactation stages, with no significant shift across development. The model for leaving during moving failed to converge due to estimation problems and could not be interpreted.

Fig. 3 Estimated marginal mean rates of infant (A) pursuing and (B) following behaviors across lactation stages in 16 mantled howler monkey mother-infant dyads (14 early, 11 middle, 7 late lactation) studied at Los Tuxtlas. Points represent model predictions from generalized linear mixed models with negative binomial distributions, back-transformed to the response scale; error bars indicate 95% confidence intervals (N=140 focal samples). * $p < 0.01$

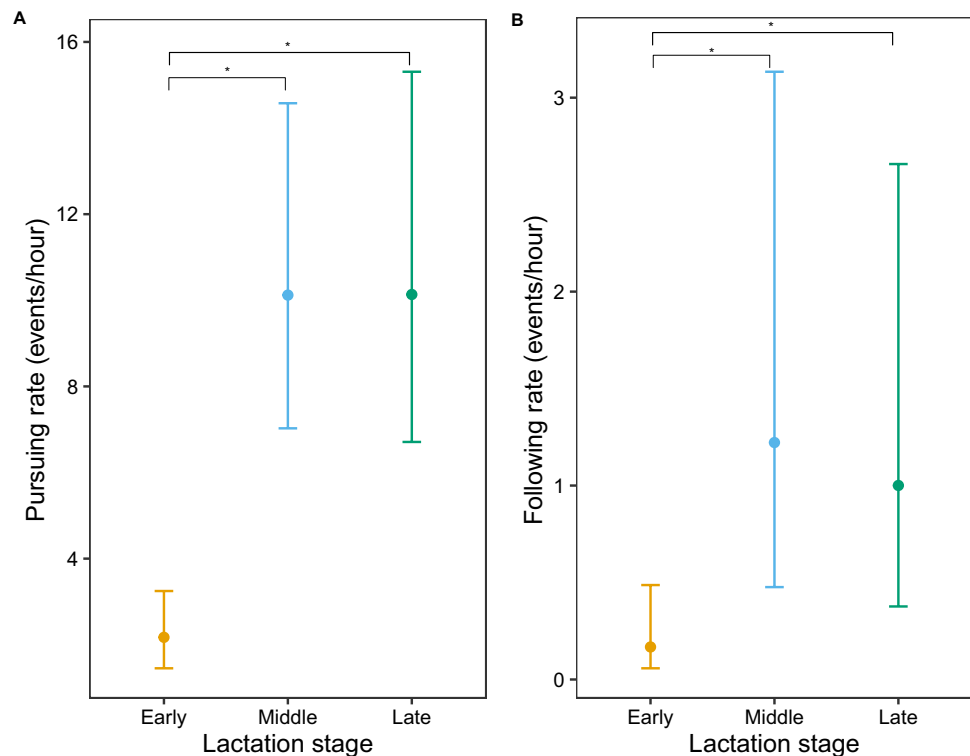


Table 2 Analysis of deviance results for proximity maintenance behaviors (P4) by maternal activity in 16 mantled howler monkey mother-infant dyads studied at Los Tuxtlas

Behavior	Activity	Actor		Lactation		Interaction	
		χ^2	p	χ^2	p	χ^2	p
Approaching	Foraging	148.4	<0.001	49.1	<0.001	5.3	0.072
	Inactive	91.5	<0.001	10.9	0.004	0.3	0.859
	Moving	3.5	0.061	51.9	<0.001	1.6	0.446
Leaving	Foraging	0.01	0.922	33.0	<0.001	1.0	0.604
	Inactive	108.2	<0.001	19.2	<0.001	0.8	0.662
	Moving	—	—	—	—	—	—
Making contact	Foraging	45.2	<0.001	18.3	<0.001	3.2	0.205
	Inactive	321.2	<0.001	21.1	<0.001	2.9	0.233
	Moving	17.0	<0.001	8.6	0.013	3.8	0.153
Breaking contact	Foraging	19.6	<0.001	19.7	<0.001	24.5	<0.001
	Inactive	120.3	<0.001	10.3	0.006	54.6	<0.001
	Moving	24.5	<0.001	10.8	0.005	8.5	0.014

All tests have $df=1$ for actor, $df=2$ for lactation and actor $\times$ lactation interaction. The leaving model during moving failed to converge and was excluded

Making contact

Offspring displayed higher rates of making contact than mothers across all activity contexts (Table 2, S11, Fig. 4C). During foraging and inactivity, offspring made contact at significantly higher rates at all lactation stages, with no significant actor $\times$ lactation interactions (Table S13). During moving, the actor effect was also significant, with offspring making more contact than mothers during middle and late lactation, but the difference was only marginal during early lactation.

Breaking contact

Breaking contact showed significant actor $\times$ lactation interactions in all three activity contexts, indicating that responsibility for breaking contact shifted across development (Table 2, S12, Fig. 4D). During foraging, offspring broke contact at higher rates than mothers during early lactation (Table S13). This difference decreased substantially by middle lactation and was absent by late lactation, indicating a shift toward shared responsibility. During inactivity, offspring consistently broke contact more than mothers across all stages.

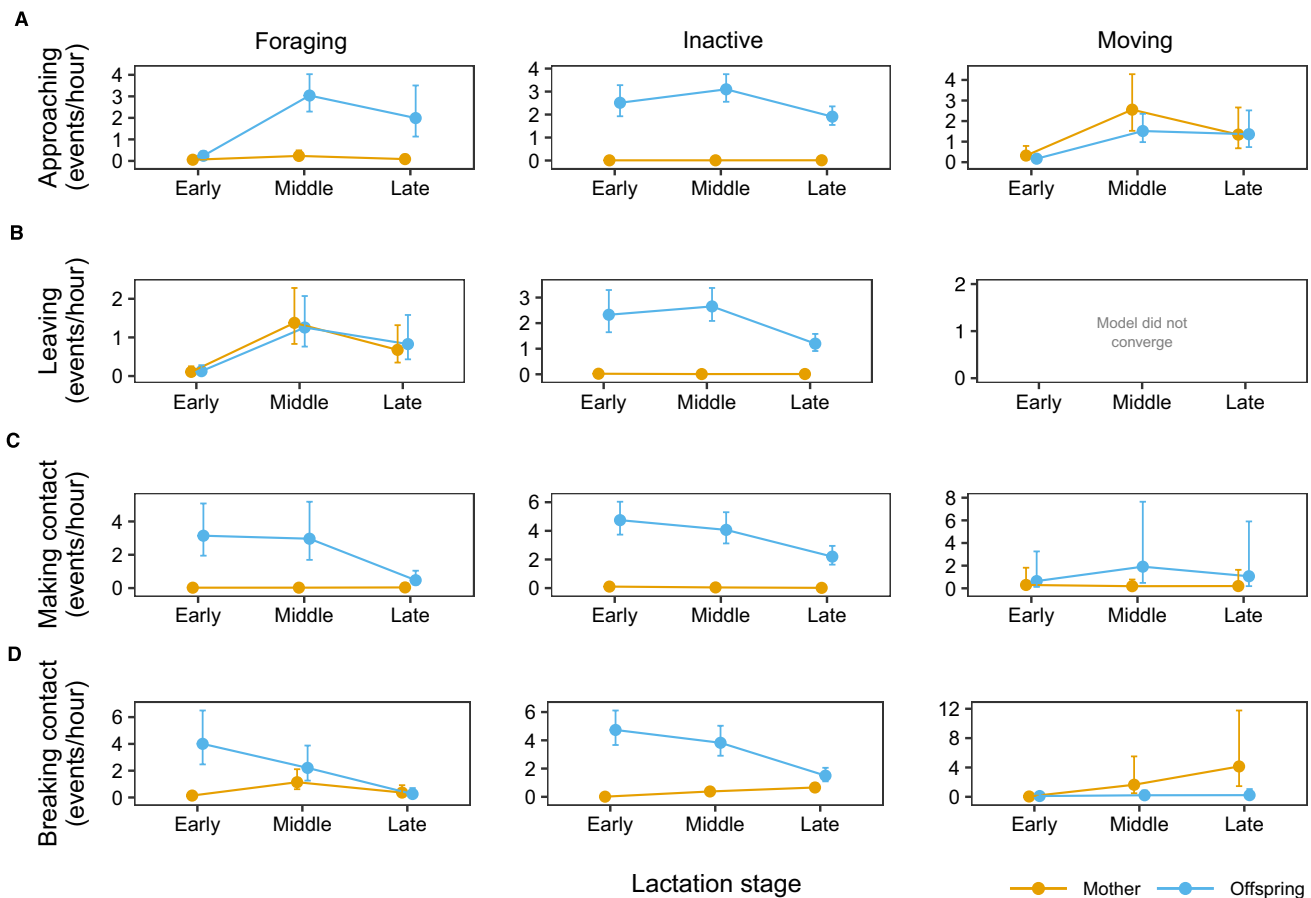


Fig. 4 Responsibility for proximity maintenance across lactation stages during different maternal activities in 16 mantled howler monkey mother-infant dyads (14 early, 11 middle, 7 late lactation) studied at Los Tuxtlas. Panels show estimated marginal mean rates of (A) approaching, (B) leaving, (C) making contact, and (D) breaking contact for mothers (orange) and offspring (blue) during foraging (left col-

umn), inactivity (middle column), and moving (right column). Points represent model predictions from generalized linear mixed models with negative binomial distributions, back-transformed to the response scale (events per hour); error bars indicate 95% confidence intervals (N=264 foraging, 280 inactive, and 278 moving observations)

However, the magnitude of this difference decreased markedly from early to late lactation, reflecting a partial shift in responsibility. During moving, the pattern reversed across development. There was no actor difference during early lactation. By middle and late lactation, mothers broke contact at significantly higher rates than offspring, indicating that mothers took primary responsibility for breaking contact during locomotion in later developmental stages.

Discussion

This study provides the first detailed characterization of parent-offspring conflict dynamics in mantled howler monkeys. Below, we evaluate each of our predictions in light of the results and discuss the broader implications for POC theory.

Regarding P1, maternal rejection generally increased through lactation, consistent with the prediction from POC theory that conflict should intensify as infants develop

independence and the cost-to-benefit ratio of continued investment shifts (Trivers 1974). However, the pattern was more nuanced than a simple monotonic increase: the trajectory of rejection depended on maternal activity context. During foraging, rejection peaked in middle lactation and then declined, whereas during inactivity, rejection increased continuously across all three stages. This activity-dependent pattern is central to our evaluation of P2.

Contrary to P2, derived from Altmann's (1980) timing hypothesis, foraging was not consistently the primary context for maternal rejection. During middle lactation, rejection was indeed highest during foraging, consistent with predictions from energy-maximizing species (Barrett et al. 1995; Mamo and Wube 2019). However, by late lactation, this pattern reversed, with rejection reaching its highest rates during inactivity. The activity-specific pattern of rejection shifted across development in ways consistent with changing opportunity costs. During middle lactation, infants are mobile but not yet nutritionally independent

(Balcells and Veà 2009); they accompany foraging mothers but cannot fully provision themselves, creating conditions where nursing and transport demands directly interfere with maternal feeding efficiency. By late lactation, infants have developed foraging competence and maintain considerable spatial independence during maternal feeding bouts (67% of time > 5 m away during foraging). With infants no longer disrupting foraging, this activity ceases to be a primary conflict context. However, infants continue soliciting nursing during rest periods, and rejection shifts to inactivity. Importantly, this shift is not simply an artifact of the greater time howler monkeys spend inactive: our models controlled for activity duration through offset terms, so the rates reflect rejection events per hour of each activity.

The prominence of conflict during maternal inactivity represents the most distinctive finding of our study. In gelada baboons, which spend up to 55% of daily time foraging, conflict peaks during feeding (Barrett et al. 1995; Mamo and Wube 2019). We propose that this pattern and ours reflect the same underlying principle viewed through the lens of different foraging strategies (Hixon 1982; Knott and DiGiorgio 2024): conflict is most acute when offspring solicitation interferes with the maternal activity that dominates time budgets and carries the highest opportunity costs. This interpretation converges with the finding that POC patterns in *A. caraya* were consistent across two populations differing substantially in density and social organization (Pavé et al. 2015), suggesting that the proximate driver of conflict expression lies in the activity-level dynamics of maternal investment rather than in broader demographic or ecological contexts. Howler monkeys possess an enlarged hindgut that enables fermentation of structural carbohydrates, but this digestive strategy requires extended retention times and favors behavioral energy minimization (Milton 1998). Given that inactivity is the dominant activity in howler monkeys, nursing during these periods may impose additional metabolic costs that compete with the physiologically demanding process of nutrient extraction from low-quality foliage (Amato et al. 2015). Moreover, howler monkey females have a higher potential lifetime reproductive output than other atelines, with estimates of 7–8 offspring per female (Raguet-Schofield and Pavé 2015), affording them greater flexibility to modulate investment in current offspring depending on environmental conditions. This flexibility may sharpen the opportunity cost of continued nursing during rest, as mothers can shift resources to future reproductive events rather than prolonging investment in current offspring whose survival probability is already established. This interpretation aligns with findings that lactating howler females increase foraging effort during early lactation but cannot fully compensate for the energetic demands of milk production (Dias et al. 2011), creating conditions where

conflict over continued nursing becomes most acute during rest. This reframing transforms an apparent contradiction of the timing hypothesis into a confirmation of its underlying logic: mothers resist offspring demands most strongly during activities critical to their own fitness, whether those activities involve acquiring resources or processing them. We acknowledge, however, that we did not directly measure the energetic costs of nursing during different activities, and future studies incorporating physiological measures would strengthen this interpretation.

Regarding P3, offspring pursuing and following behaviors increased between early and middle lactation as predicted, but contrary to prediction, these behaviors plateaued rather than decreased in late lactation. This increase between early and middle lactation coincides with the development of locomotor independence (Balcells and Veà 2009; Raguet-Schofield and Pavé 2015) and with the period of highest maternal rejection during foraging, suggesting that infants adjusted their solicitation in response to changing maternal behavior and their own increasing mobility. The plateau in late lactation—rather than a decrease—indicates that infants maintained elevated solicitation levels even as they approached nutritional independence.

The absence of further escalation of conflict in late lactation is inconsistent with scramble competition models, which predict continuous increases in offspring demands (Parker and Macnair 1979). However, we recognize that our behavioral measures—pursuing and following—capture locomotor behavior that may reflect multiple motivations beyond signaling need. Pursuing a moving mother does not necessarily constitute a costly signal in the sense proposed by signaling theory (Godfray 1991; Bateson 1994), as the mother may not perceive following behavior as a solicitation signal, and these behaviors lack the conspicuous costly components (such as loud vocalizations) typically associated with honest signals of need. Nonetheless, the interpretation of offspring solicitation as potentially honest is not without precedent in *Alouatta*: Pavé et al. (2010) proposed that *A. caraya* mothers may use infant distress calls as honest signals of need, and Pavé et al. (2015) found that infant distress was often decoupled from maternal rejection, occurring instead during group displacements when infants were forced to travel independently; contexts suggesting genuine need rather than manipulative escalation. The developmental modulation we observed may therefore be better interpreted as reflecting the interplay between increasing locomotor competence and changing nutritional needs across development, rather than as direct evidence for or against specific signaling frameworks. Distinguishing between these interpretations would require concurrent measurement of vocal solicitation and behavioral indicators of offspring condition.

Supporting P4, responsibility for proximity maintenance rested primarily with infants across all lactation stages and activity contexts. Infants were responsible for approaching mothers, initiating contact, and breaking contact. This pattern resembles wild blue monkeys (*Cercopithecus mitis*; Förster and Cords 2004) and Geoffroy's spider monkeys (*Ateles geoffroyi*), where infants were responsible for regulating proximity in over 85% of observed dyad-months (Arbaiza-Bayona et al. 2022), but contrasts with cercopithecines where mothers actively restrain infants (Maestripieri 1994). Notably, a similar pattern of infant-driven proximity has been reported for *A. caraya*, where infants were primarily responsible for maintaining proximity to mothers (Pavé et al. 2015). The absence of maternal restraint likely reflects the low-risk social environment of howler groups, characterized by small group size, infrequent aggression, minimal infant handling by non-mothers (Czerwinski et al. 2025b), and a permissive maternal style that may also serve to promote infant independence (Czerwinski et al. 2025a). The finding that mothers increasingly took responsibility for breaking contact during locomotion in later developmental stages likely reflects the practical need for mothers to initiate travel independently as infants become spatially autonomous. We note that proximity maintenance is often quantified using the Hinde index (Hinde and Atkinson 1970) or the Brown-modified index (Brown 2001; see Scott et al. 2023 for a recent review). However, because the Hinde index collapses approaching, leaving, making contact, and breaking contact into a single value that can produce identical scores from fundamentally different behavioral patterns (Brown 2001), we opted for the more informative approach of modeling each behavioral component separately within each activity context.

Maternal rejection dynamics showed both convergence and divergence with patterns documented in the congener *A. caraya*. Rejection increased significantly between early and middle lactation, consistent with intensifying conflict as infants develop independence. Coincidentally, Pavé et al. (2015) recorded the highest rejection rates between months 3 and 7, a period similar to our middle lactation stage (4–8 months), suggesting comparable conflict timing during infancy across these two howler species. Notably, Pavé et al. (2010) reported that maternal rejection in *A. caraya* occurred during feeding, resting, and nursing of other infants, providing cross-species support for the finding that conflict in howler monkeys is not confined to foraging contexts but extends prominently to periods of inactivity. The onset of rejection in mantled howlers (middle lactation, ~4–8 months) falls between the early rejection documented in *A. caraya* (month 2; Pavé et al. 2010, though based on only two infants in the first months of life) and the much later onset in Geoffroy's spider monkeys

(19–21 months; Arbaiza-Bayona et al. 2022). This variation in conflict timing among platyrrhines is likely related to differences in life history variables between genera, including interbirth intervals and age at weaning (Raguét-Schofield and Pavé 2015).

Several limitations warrant acknowledgment. Our sample of 16 dyads provided adequate power for main effects but limited ability to detect subtle interactions, particularly in late lactation where only 7 dyads contributed data. The stage-based categorization, while enabling comparison across broad developmental periods, may have obscured finer temporal dynamics documented in *A. caraya* (Pavé et al. 2010). Our sampling effort of approximately 4–5 days per dyad per lactation stage, while comparable to other field studies of howler monkey behavior, may have limited our ability to capture rare events or short-term fluctuations in conflict intensity. Additionally, we did not control for maternal parity, infant sex, or food availability, factors shown to influence POC expression in *A. caraya* and maternal behavior in mantled howler monkeys (Pavé et al. 2015; de la Torre et al. 2021). Future studies should incorporate larger samples, finer temporal resolution, explicit ecological measurements, and concurrent recording of vocal solicitation to better characterize the signaling dynamics of parent–offspring interactions.

In conclusion, this study makes three contributions to understanding parent–offspring conflict. First, we provide empirical evidence that POC in mantled howler monkeys follows developmental trajectories broadly consistent with theoretical predictions, extending the taxonomic scope of POC research to an understudied platyrrhine species. Second, we demonstrate that the activity context of conflict expression depends critically on species-specific foraging strategy, with inactivity replacing foraging as the primary conflict arena in this time-minimizing species—a finding that refines the timing hypothesis for application across diverse foraging strategies. Third, the developmental modulation of offspring solicitation and the infant-driven pattern of proximity maintenance add to a growing body of evidence that parent–offspring dynamics in platyrrhines differ in important ways from the cercopithecine-dominated literature. Together, these findings underscore that testing evolutionary theory across ecologically diverse taxa reveals both the generality of core principles and the necessity of incorporating ecological context into behavioral predictions.

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Author contributions ARN and PADD originally formulated the idea; MC and PADD developed methodology; MC conducted fieldwork; ARN and PADD provided project administration; MC and PADD performed statistical analyses; all authors wrote the manuscript.

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Data availability The datasets used in this study are available at https://figshare.com/s/ac26195a1e82323271cf (https://figshare.com/s/ac26195a1e82323271cf).

Declarations

Ethics approval This study was noninvasive and adhered to the ASP Code of Best Practices in Field Primatology and the ASP Principles for the Ethical Treatment of Nonhuman Primates. Research protocols were approved by the Secretaría de Medio Ambiente y Recursos Naturales and complied with the legal requirements of Mexican law (NOM 059-SEMARNAT-2010; SEMARNAT SGPA/DGVS/04015/21 and SGPA/DGVS/00278/22).

Conflicts of interest The authors declare no competing interests.

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